PROPERTY (BETA), DECENCY AND REASONABLENESS ARE PRESERVED UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4024-4036, immediately following the statement at 4018-4022.

Editorial note. Two same-chapter line-indexed stubs are recorded in assembly.external_references; no cross-chapter pointers and no bibliography dependence (check_ref_load criterion: nothing discarded).

Allowed standard prerequisites (editorial).

- Base change case of the relative conditions. Lemma lemma-base-change-relative-conditions of this chapter, cited at proof lines 4028-4029. Reduces 'T x_Z X has (omega) for all T -> Z' to the case where the middle object itself carries (omega): X -> Y with Y having (omega) gives X has (omega).
- Transitivity over a space with the property. Lemma lemma-property-over-property of this chapter, cited at proof lines 4034-4035. Supplies the reduced claim: Y has (omega) and f : X -> Y has (omega) imply X has (omega).

Source reference. Original lemma 03L0: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3953–3953. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ABY: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3963–3963. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Having property (β), being decent, or being reasonable is preserved under compositions.*

Proof. Let $\omega \in \{\beta, decent, reasonable\}$. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of algebraic spaces over the scheme S. Assume f and g both have property (ω). Then we have to show that for any scheme T and morphism $T \to Z$ the space $T \times_Z X$ has (ω). By Lemma 03L0 this reduces us to the following claim: Suppose that Y is an algebraic space having property (ω), and that $f : X \to Y$ is a morphism with (ω). Then X has (ω). This is the content of Lemma 0ABY. □

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.